

Linear Regression – Full Notes (Beginner → Intermediate)

Linear Regression is a Supervised Machine Learning Algorithm used to predict continuous numerical values.

Examples:

- House Price Prediction
- Salary Prediction
- Temperature Forecast
- Sales Prediction

1. What is Linear Regression?

Linear Regression finds the linear relationship between input and output.

Example:

Area (x)	Price (y)
1000	200
1500	300
2000	400

As area increases, price increases.

When plotted on a graph, it forms a straight line.

Formula:

$$y = mx + c$$

Where:

- y → Predicted value
- x → Input feature
- m → Slope
- c → Intercept

2. Simple Linear Regression

Used when there is only one feature.

Formula:

$$y = mx + c$$

Example:

$$y = 2x + 5$$

If:

$$x = 10$$

Then:

$$y = 2(10) + 5$$

$$y = 25$$

3. Slope Formula

$$m = (n\sum xy - (\sum x)(\sum y)) / (n\sum x^2 - (\sum x)^2)$$

Where:

- n = Number of samples
- $\sum x$ = Sum of x values
- $\sum y$ = Sum of y values

here:

$$n = 3$$

$$x = [1000, 1500, 2000]$$

$$y = [200, 300, 400]$$

$$n=3$$

$$\sum x = 1000 + 1500 + 2000 = 4500$$

$$\sum y = 200 + 300 + 400 = 900$$

$$\sum xy = (1000 * 200) + (1500 * 300) + (2000 * 400)$$

$$\sum xy = 200000 + 450000 + 800000 = 1450000$$

$$\sum x^2 = 1000^2 + 1500^2 + 2000^2$$

$$\sum x^2 = 1000000 + 2250000 + 4000000 = 7250000$$

Slope Formula:

$$m = (n\sum xy - (\sum x)(\sum y)) / (n\sum x^2 - (\sum x)^2)$$

$$m = (3 * 1450000 - 4500 * 900) / (3 * 7250000 - 4500^2)$$

$$m = (4350000 - 4050000) / (21750000 - 20250000)$$

$$m = 300000 / 1500000$$

$$m = 0.2$$

4. Intercept Formula

$$c = (\sum y - m\sum x) / n$$

$$c = (900 - 0.2 * 4500) / 3$$

$$c = (900 - 900) / 3$$

$$c = 0$$

5. Prediction Formula

$$\hat{y} = mx + c$$

$$\hat{y} = 0.2 * 1000 + 0$$

$$\hat{y} = 200$$

$\hat{y}$ = Predicted output

6. Cost Function

Used to calculate error.

Mean Squared Error (MSE):

$$J(m,c) = (1/n) \sum (y - \hat{y})^2$$

Where:

- y = Actual value
- $\hat{y}$ = Predicted value

Goal: Minimize the error.

$$J(m,c) = (1/n) \sum (y - \hat{y})^2$$

$$J(0.2, 0) = (1/3) \sum (y - \hat{y})^2$$

$$(200-200)^2 + (300-300)^2 + (400-400)^2 = 0$$

x	y	$\hat{y}$
1000	200	200
1500	300	300
2000	400	400

Cost Function:

$$J(0.2,0) = 1/3[(200-200)^2 + (300-300)^2 + (400-400)^2]$$

7. MAE (Mean Absolute Error)

$$MAE = (1/n) \sum |y - \hat{y}|$$

Example:

Actual: [10, 20]

Predicted: [12, 18]

Difference: $|10-12| = 2$

$|20-18| = 2$

$MAE = (2 + 2) / 2 = 2$

$MAE = 2$

Lower MAE = Better model

8. MSE (Mean Squared Error)

$MSE = (1/n) \sum (y - \hat{y})^2$

Example:

Error: [2, -2]

Squared: [4, 4]

$MSE = 4$

9. RMSE

$RMSE = \sqrt{MSE}$

$\sqrt{4} = 2$

Advantage: Uses the same unit as the target variable.

10. R^2 Score

Measures model performance.

Formula:

$R^2 = 1 - [\sum (y - \hat{y})^2 / \sum (y - \bar{y})^2]$

$R^2 = 1 - [200 / 300]$

$R^2 = 1 - 0.6667$

$R^2 = 0.3333$

Range:

1 → Perfect 0 → Average < 0 → Poor

11. Multiple Linear Regression

Used with multiple features.

Formula:

$$y = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n$$

Example:

$$\text{Price} = 5000 + 2000(\text{Age}) + 300(\text{Size})$$

lets

Age = 10 years

Size = 1000 sq ft

so

$$\text{Price} = 5000 + 2000(10) + 300(1000)$$

$$\text{Price} = 5000 + 20000 + 300000$$

$$\text{Price} = 325000$$

12. Vector Form

$$Y = X\beta + \epsilon$$

Where:

- Y = Target
- X = Feature Matrix
- β = Coefficients
- ϵ = Error

Example:

$$Y = [[200], [300], [400]]$$

$$X = [[1000, 1], [1500, 1], [2000, 1]]$$

$$\beta = [[0.2], [0]]$$

$$\epsilon = [[0], [0], [0]]$$

$$Y = [200, 300, 400]$$

$$X = [[1000, 1], [1500, 1], [2000, 1]]$$

$$\beta = [0.20]$$

$$\epsilon = [0, 0, 0]$$

Multiple Linear Regression of matrix formula:

$$Y = X\beta + \epsilon$$

$$\epsilon = 0$$

$$Y = X\beta$$

$$Y = [[1000, 1], [1500, 1], [2000, 1]] * [[0.2], [0]]$$

1st row:

$$1000 * 0.2 + 1 * 0 = 200$$

2nd row:

$$1500 * 0.2 + 1 * 0 = 300$$

3rd row:

$$2000 * 0.2 + 1 * 0 = 400$$

So:

$$Y = [200, 300, 400]$$

so your actual Y matches perfectly.

That means:

$$\text{Coefficient} = 0.2$$

$$\text{Intercept} = 0$$

$$\text{Error } (\epsilon) = 0$$

$$\text{Model is perfectly fit } (R^2 = 1)$$

relation is:

$$y = 0.2x$$

That means for every 1 unit increase in x, y increases by 0.2 units.

13. Normal Equation

Directly calculates coefficients.

Formula:

$$\beta = (X^T X)^{-1} X^T y$$

Meaning:

- X = Input Matrix
- X^T = Transpose
- $(X^T X)^{-1}$ = Matrix Inverse
- y = Output
- β = Weights

Example:

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$$

$$y = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

Result:

$$\beta = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

Equation:

$$y = 2x$$

14. Gradient Descent

Iterative optimization.

Update:

$$m = m - \alpha(\partial J/\partial m)$$

$$c = c - \alpha(\partial J/\partial c)$$

Where: α = Learning Rate

Large $\alpha \rightarrow$ Overshoot

Small $\alpha \rightarrow$ Slow

15. Assumptions of Linear Regression

- Linearity
- Independence
- Homoscedasticity
- Normality
- No Multicollinearity

16. Train Test Split

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split( X, y,
test_size=0.2, random_state=42 )
80% Train
20% Test
```

17. Build Linear Regression

```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit( X_train, y_train )
```

18. Prediction

```
y_pred = model.predict(X_test)
```

19. Evaluation

```
from sklearn.metrics import (mean_absolute_error,
```

```
mean_squared_error, r2_score )
```

20. Coefficients

```
print("Coefficients:", model.coef_)  
print("Intercept:", model.intercept_)
```

21. Visualization

```
import seaborn as sns  
import matplotlib.pyplot as plt  
sns.scatterplot( x=y_test, y=y_pred )  
plt.show()
```

22. Advantages

- ✓ Easy
- ✓ Fast
- ✓ Interpretable
- ✓ Works well for linear data

23. Disadvantages

- x Sensitive to outliers
- x Poor for nonlinear data
- x Overfitting possible

24. Real Project Flow

Load Dataset ↓

Cleaning ↓

Split ↓

Train ↓

Predict ↓

Evaluate ↓

Deploy

Short Summary

Linear Equation:

$y = mx + c$

Multiple:

$y = b_0 + b_1x_1 + \dots$

Normal Equation:

$$\beta = (X^T X)^{-1} X^T y$$

Metrics:

MAE

MSE

RMSE

R^2